Li et al. Journal of Translational Medicine (2025) 23:117
https://doi.org/10.1186/s12967-025-06140-y

Journal of
Translational Medicine

RESEARCH

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# Elevated LIF and JAK-STAT activation drive severe COVID-19 in myeloma patients receiving the BCMA-CD3 bispecific antibody Elranatamab

Ziping Li[1], Fujing Zhang[1], Xianghong Jin[1] and Junling Zhuang[1*]

## Abstract

**Background** Immunotherapy is a significant risk factor for severe COVID-19 in multiple myeloma (MM) patients. Understanding how immunotherapies lead to severe COVID-19 is crucial for improving patient outcomes.

**Methods** Human protein microarrays were used to examine the expression of 440 protein molecules in MM patients treated with bispecific T-cell engagers (BiTe) (n = 9), anti-CD38 monoclonal antibodies (mAbs) (n = 10), and proteasome inhibitor (PI)-based regimens (n = 10). Differentially expressed proteins (DEPs) were identified and analyzed using bioinformatics.

**Results** BiTe therapy was associated with a higher incidence of severe COVID-19. We identified 21 and 29 DEPs between BiTe and anti-CD38 mAbs group, and BiTe and PI-based group, respectively, along with 25 DEPs between the anti-CD38 and PI groups. Principal component analysis and clustering showed distinct protein expression profiles between the BiTe and PI groups. Gene Ontology (GO) analysis revealed that DEPs between the BiTe and PI groups were related to cytokine activity and leukocyte migration. Kyoto Encyclopedia of Genes and Genomes (KEGG) analysis showed that these DEPs were enriched in cytokine-cytokine receptor interaction and JAK-STAT signaling pathways. Leukemia inhibitory factor (LIF) is the most correlated with other DEPs and thus may play a key role in both enriched pathways, and the level of LIF protein was highest in the BiTe group.

**Conclusions** BiTe therapy is linked to a higher risk of severe COVID-19 due to an inflammatory cytokine storm, with LIF and the JAK-STAT pathway playing key roles. Targeting LIF and JAK-STAT pathway may help reduce severe COVID-19 in MM patients treated with BiTe.

**Keywords** Multiple myeloma, Immune therapy, Bispecific T-cell engagers, Severe COVID-19, Human protein microarray

## Background

Multiple Myeloma (MM) is a hematologic malignancy caused by aberrant plasma cell proliferation in the bone marrow, which remains incurable [1]. In recent years, the treatment of MM patients has been revolutionized through the use of targeted immune therapies, including anti-CD38 monoclonal antibodies (mAbs), bispecific T-cell engagers (BiTe), and chimeric antigen receptor T cells (CAR T cells) [1]. Targeted immune therapies, while effective in controlling myeloma, exhibit distinct immunomodulatory effects such as immunosuppression, immune overactivation, or inadequate immune response, which may influence patient susceptibility to infections, including the novel coronavirus disease 2019 (COVID-19) [2].

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During the COVID-19 pandemic, MM patients are highly susceptible to infections due to both the disease itself and the intensive treatments they undergo. Various studies have shed light on BiTe therapy targeting B cell maturation antigen (BCMA) is significantly associated with severe COVID-19, and the impact of anti-CD38 mAbs on hospitalization and mortality after COVID-19 infection in MM patients is controversial [3, 4]. Our early studies have found that targeted immunotherapy, including BiTe therapy targeting BCMA and anti-CD38 mAbs, is an independent risk factor for the development of severe COVID-19 in MM patients. And patients treated with BiTe had the highest probability of developing severe COVID-19. However, detailed comparative analyses of effects on COVID-19 severity among different MM regimens remain scarce. The specific molecular mechanism by which targeted immunotherapies, especially BiTe therapy, trigger severe COVID-19 in MM patients is still unclear, and few studies have been conducted.

Understanding the underlying mechanisms by which targeted immunotherapies, particularly BiTe therapies, cause severe COVID-19 is critical for optimizing treatment strategies and improving patient outcomes. This study aims to elucidate the impact of different anti-MM regimens on COVID-19 severity and delve into the mechanisms and prognostic molecules related to immunotherapy triggered severe COVID-19. Protein microarrays were used to detect the expression of 440 protein molecules in MM patients treated with BiTe, anti-CD38 mAbs, and proteasome inhibitor (PI)-based regimens. These protein molecules are mainly involved in immune response, inflammatory response, coagulation pathway, and vascular endothelial generation. We further performed a comprehensive analysis of differentially expressed proteins (DEPs) between these treatment groups using bioinformatics methods. These findings provided important insights into the mechanisms underlying the development of severe COVID-19 in MM patients receiving targeted immunotherapy and laid the foundation for developing potential prognostic markers and therapeutic targets.

## Methods

### Study design and participants

The study included three groups of relapsed/refractory MM (R/R MM) patients infected with COVID-19: nine patients treated with BiTe (Group 1, the BiTe involved in our study is Elranatamab, a humanized BCMA-CD3 bispecific antibody. Pfizer, Inc.); ten R/RMM patients received anti-CD38 mAbs (Group 2, the anti-CD38 mAbs involved in our study is daratumumab. Janssen-Cilag, Inc.); ten R/RMM patients received PI-based therapy (Group 3). Among the 10 patients in group 3, 4 patients received bortezomib (Janssen-Cilag, Inc.) as the proteasome inhibitor-based therapy, 4 patients received carfilzomib (Onyx Pharmaceuticals, Inc.), and 2 patients received ixazomib (Takeda Pharma A/S, Inc.). All MM patients included in our study were not previously infected with COVID-19. Except for the anti-myeloma regimen, the three groups of patients represented comparable demographic and clinical characteristics. Participants' clinical data is listed in Table 1. Plasma samples were collected at 7–10 days after infections. The criteria for severity of COVID-19 were defined according to the diagnostic criteria [5]. Active MM was defined as disease requiring induction treatment rather than maintenance therapy (stable disease) according to the latest research on the diagnosis and treatment of multiple myeloma [6]. All subjects were recruited at Peking Union Medical College Hospital, Beijing, China. This study was approved by the local Institutional Medical Ethics Committee (No. I-23PJ229). Written informed consent was obtained from each patient.

### Antibody array assay

Plasma obtained from the participants was used to research specific biomarkers by Human Cytokine Antibody Array (GSH-CAA-440, RayBiotech Company, Norcross, GA, USA), according to the manufacturer's instructions. This antibody array simultaneously detects 440 cytokines in a single experiment by utilizing a sandwich technique with 440 antibody dots arranged in four duplicates printed onto the glass. Briefly, plasma samples were diluted (1:2) and added into the array pools to incubate with capture antibodies overnight. After washing, the arrays were incubated with a biotin-conjugated anti-cytokine antibody mix for 2 h at room temperature. Cy3-conjugated streptavidin was added to bind with biotin from the detection antibodies and the fluorescent signal was detected using an InnoScan 300 Microarray Scanner (Innopsys, France). Signal values were captured with Mapix software. The data was normalized using positive control values from the array with the RayBiotech analysis tool (GenePix, ScanArray Express, ArrayVision, MicroVigene, etc.), specifically designed to analyze the data of Human Cytokine Antibody Array G series 440 with Microsoft Excel technology.

### Statistical and Bio-informatics analysis

Categorical variables were described as frequency and percentages. Normally distributed continuous variables were presented as means (± standard deviation [SD]), whereas non-normally distributed data were presented as medians with interquartile range (IQR). Proportions for categorical variables were compared using the $\chi 2$ test, and the Fisher exact test was used when data were

**Table 1** Participant characteristics for antibody array

| Characteristics | Group 1<br>n = 9 | Group 2<br>n = 10 | Group 3<br>n = 10 | P value |
|---|---|---|---|---|
| Age (y), median (IQR) | 67 (62–74) | 67 (64–70.5) | 70.5 (63.5–73) | 0.785 |
| Gender (M/F), n (%) | | | | 0.645 |
| Male (M) | 5 (55.6) | 6 (60.0) | 4 (40.0) | |
| Female (F) | 4 (44.4) | 4 (40.0) | 6 (60.0) | |
| Immunoglobulin subtypes, n (%) | | | | 0.821 |
| IgG | 5 (55.6) | 7 (70.0) | 8 (80.0) | |
| IgA | 2 (22.2) | 1 (10.0) | 1 (10.0) | |
| IgD | 0 (0) | 1 (10.0) | 0 (0) | |
| LC | 2 (22.2) | 1 (10.0) | 1 (10.0) | |
| R-ISS stage, n (%) | | | | 1.000 |
| I | 2 (22.2) | 2 (20.0) | 2 (20.0) | |
| II | 5 (55.6) | 5 (50.0) | 6 (60.0) | |
| III | 1 (11.1) | 2 (20.0) | 1 (10.0) | |
| Unknown | 1 (11.1) | 1 (10.0) | 1 (10.0) | |
| Auto-HSCT, n (%) | 1 (11.1) | 2 (20.0) | 2 (20.0) | 1.000 |
| MM status at COVID-19, n (%) | | | | 1.000 |
| Inactive | 5 (55.6) | 6 (60.0) | 6 (60.0) | |
| Active | 4 (44.4) | 4 (40.0) | 4 (40.0) | |
| Comorbidities, n (%) | | | | 1.000 |
| 0 | 2 (22.2) | 2 (20.0) | 3 (30.0) | |
| ≥ 1 | 7 (77.8) | 8 (80.0) | 7 (70.0) | |
| Vaccination, n (%) | 2 (22.2) | 2 (20.0) | 4 (40.0) | 0.668 |
| Time intervals from the last treatment to COVID-19 onset (days), median (range) | 6 (2, 10) | 4.5 (1,10) | 4.5 (2, 10) | 0.403 |
| Severity of COVID-19, n (%) | | | | 0.040 |
| Severe | 6 (66.7) | 3 (30.0) | 1 (10.0) | |
| Non-severe | 3 (33.3) | 7 (70.0) | 9 (90.0) | |
| LOT, median (IQR) | 3 (3, 4) | 2 (2, 2.75) | 2 (2, 3) | 0.067 |
| Treatments | Elranatamab (9 patients) | DPd (5 patients)<br>DRd (3 patients)<br>Dd (2 patients) | KPd (4 patients)<br>VPd (2 patients)<br>VRd (2 patients)<br>IR (2 patients) | – |

Group 1: MM patients treated with bispecific T-cell engagers therapy; Group 2: MM patients treated with anti-CD38 mAbs therapy; Group 3: MM patients treated with PI-based therapy. Abbreviations. MM: multiple myeloma, IQR: interquartile range, R-ISS: Revised International Staging System, Auto-HSCT: autologous hematopoietic stem cell transplantation, LOT: lines of treatment, D: Daratumumab, P: Pomalidomide, K: Carfilzomib, V: Bortezomib, R: Lenalidomide, I: Ixazomib, d: dexamethasone

limited. The quantized variables of parameters were tested by t-test. Nonparametric variables were tested by Mann Whitney U test or Kruskal–Wallis test. All statistical analysis was performed using SPSS v.27.0 (IBM Corp., Armonk, NY, USA). A two-tailed P value < 0.05 was considered statistically significant.

The R software package "limma" from R/Bioconductor was used to analyze differentially expressed proteins (DEPs) in MM patients from different treatment groups. This software was specifically used to compare the levels of factors between two different groups using moderated t-statistics. The results included (log2) fold changes and p values for each factor. False discovery rate (FDR) (the Benjamini–Hochberg method) was used to calculate adjusted P values (p adj.). Protein expression values of the |log2 FC| > 1.5 and p value or adjusted p-value < 0.05 were significant. DEPs were visualized in a volcano plot (R package "ggplot2"). Principal component analysis (PCA) (R package "ggfortify"), cluster heatmap analysis (R package "gplots"), Gene Ontology (GO) enrichment analysis (R package "clusterProfiler"), Kyoto Encyclopedia of Genes and Genomes (KEGG) enrichment analysis (R package "clusterProfiler") and Friends analysis (R package

"GOSemSim") were executed using the open-source program R (version 4.3.3).

## Results

### Patient demographics

Demographics and clinical information of the three groups of R/R MM patients were summarized in Table 1. Age, gender, paraprotein subtypes, revised-international staging system (R-ISS), autologous hematopoietic stem cell transplantation (auto-HSCT) status, MM status at COVID-19, comorbidities, time intervals from the last treatment to COVID-19 onset, and COVID-19 vaccination were matched in three groups without statistical differences. In each group, the utilization rate of auto-HSCT was low mainly because many patients were over 70 years old or had impaired cardiac or renal functions. Six in nine (6/9, 66.7%) patients treated with BiTe (Group 1) developed severe COVID-19, while 3/10 (30%) patients received anti-CD38 mAbs (Group 2) and 1/10 (10%) patients received PI-based therapy (Group 3) experienced severe COVID-19. There was a significant statistical difference in the occurrence of severe COVID-19 among the three groups (p = 0.040). Further analyses revealed that patients treated with BiTe (Group 1) had a significantly higher rate of severe COVID-19 than those with PI-based therapy (Group 3) (p = 0.019, Fig. 1), but not between Group 1 versus Group 2 (p = 0.179, Fig. 1) nor between Group 2 versus Group 3 (p = 0.582, Fig. 1). Regarding the COVID—19 outcomes, six patients died and five were admitted to the ICU due to COVID-19 in Group 1; one patient died and two were admitted to the ICU due to COVID-19 in Group 2; one patient was admitted to the ICU and no patient died due to COVID-19 in Group 3. Additionally, all patients with severe COVID-19 across the three groups received Paxlovid and dexamethasone, and patients admitted to the ICU also received mechanical ventilation. In Group 1, all non-severe COVID-19 patients were treated with traditional Chinese medicine and antipyretics. In Group 2, two non-severe COVID-19 patients took only antipyretics, and the rest were treated with traditional Chinese medicine combined with antipyretics. In Group 3, one non-severe COVID-19 patient used only antipyretics, while the others were treated with the combination of traditional Chinese medicine and antipyretics.

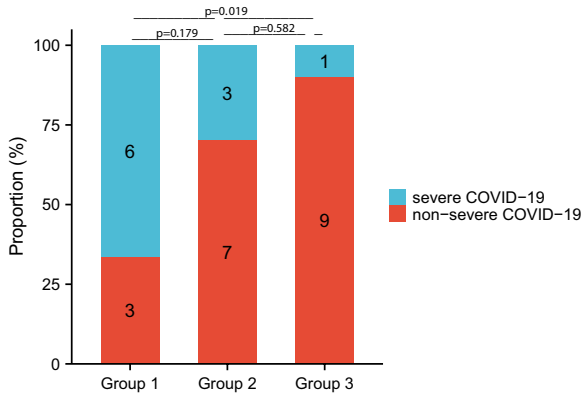


**Fig. 1** Incidence of severe COVID-19 among three treatment groups in MM patients. Group 1: MM patients treated with BCMA-CD3 bispecific antibody therapy; Group 2: MM patients treated with anti-CD38 mAbs therapy; Group 3: MM patients treated with PI-based therapy. Abbreviations. MM: multiple myeloma

### Identification of differential expression proteins (DEPs)

To determine the potential mechanism of different anti-MM regimens on COVID-19 severity, we performed protein microarray testing in three groups of MM patients, mainly involving proteins related to immunity, inflammation, coagulation, and vascular endothelium. Consequently, a total of 21 DEPs were detected in patients receiving BiTe therapy compared to those treated with anti-CD38 mAbs (Fig. 2, A), 29 DEPs were identified between patients treated with BiTe and PI-based therapy (Fig. 2, B), and 25 DEPs were found between patients receiving anti-CD38 mAbs and those receiving PI-based therapy (Fig. 2, C). These results were presented in the form of volcano plots (Fig. 2).

### Principal component analysis of DEPs

The PCA is conducted on all DEPs among three groups. The first two principal components (PC) are plotted to show the difference between the two groups. Samples within the same treatment group cluster together (Fig. 3), showing good repeatability. There was a small overlap between the BiTe group and the anti-CD38 mAbs group (Fig. 3A), as well as between the anti-CD38 mAbs group and PI-based group (Fig. 3C), but a clear separation was observed between the BiTe group and the PI-based group (Fig. 3B). The PCA results showed that different anti-MM regimens caused varied protein expression.

### Hierarchical clustering analysis of DEPs

We performed a cluster analysis between the three groups, and the results were presented as heatmaps. Euclidean distance and complete cluster are used to analyze dissimilarities between two groups. There was no sample cross cluster together between the BiTe group and PI-based group, while 2 and 3 samples were cross clustered together between the BiTe group and anti-CD38 mAbs group, and between the anti-CD38 group mAbs and PI-based group, respectively (Fig. 4). Overall, compared with patients treated with anti-CD38 mAbs, FABP2, HGF, IL–18, MCP–3, MIG, Syndecan–1, E–Selectin, PDGF Rb, TECK, IGF–1R, and MIP–3a were

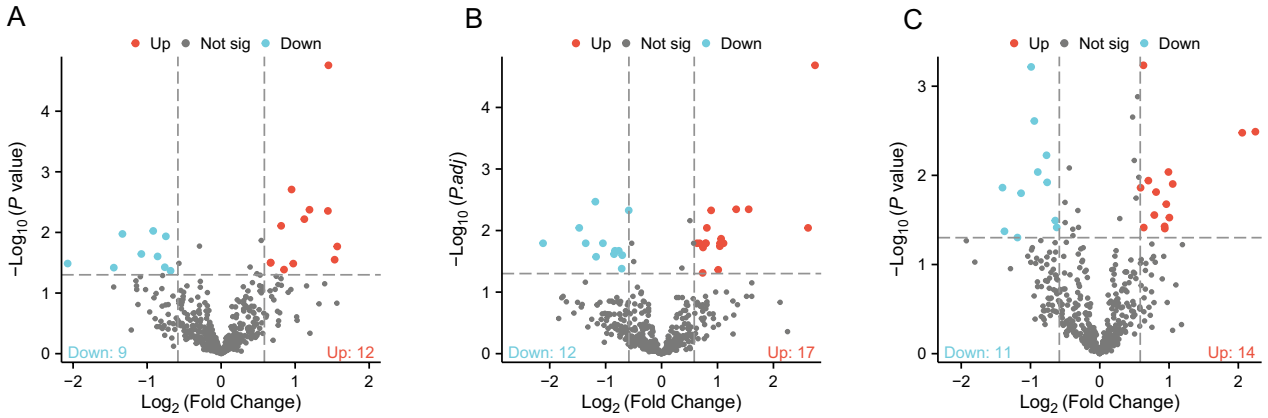


**Fig. 2** Differential expression proteins (DEPs) among three treatment groups in MM patients. Group 1: MM patients treated with BCMA-CD3 bispecific antibody therapy; Group 2: MM patients treated with anti-CD38 mAbs therapy; Group 3: MM patients treated with PI-based therapy. **A** DEPs between MM patients treated with BiTe therapy and anti-CD38 mAbs therapy. **B** DEPs between MM patients treated with BiTe therapy and PI-based therapy. **C**. DEPs between MM patients treated with anti-CD38 mAbs therapy and PI-based therapy. Red presents up regulation, blue presents down regulation, and grey presents no difference. Abbreviations. MM: multiple myeloma

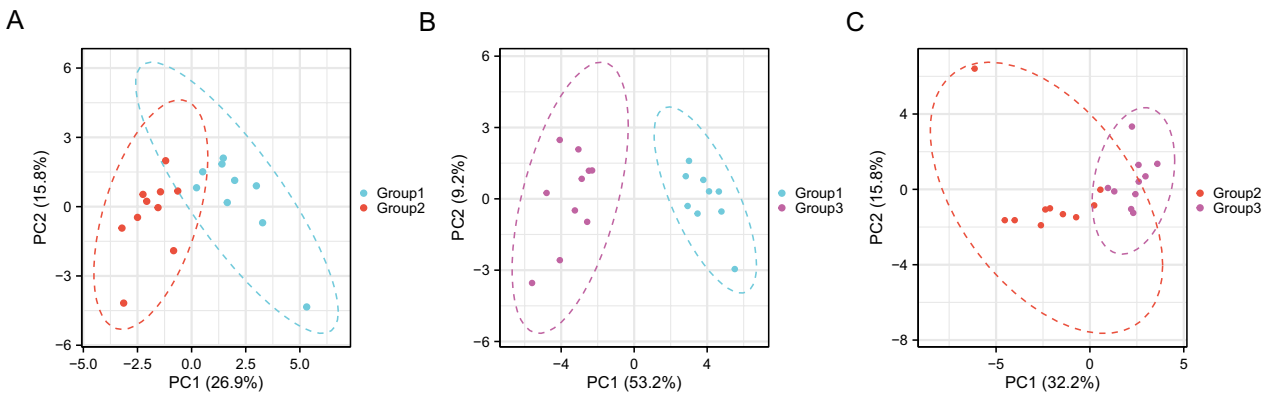


**Fig. 3** Principal component analysis (PCA) of DEPs among three treatment groups in MM patients. The first two principal components (PC) are plotted to show the difference between two groups. Euclidean distance and complete cluster are used to analyze dissimilarities between two groups. Group 1: MM patients treated with BCMA-CD3 bispecific antibody therapy; Group 2: MM patients treated with anti-CD38 mAbs therapy; Group 3: MM patients treated with PI-based therapy. **A** PCA result between MM patients treated with BiTe therapy and anti-CD38 mAbs therapy. **B** PCA result between MM patients treated with BiTe therapy and PI-based therapy. **C** PCA result between MM patients treated with anti-CD38 mAbs therapy and PI-based therapy. DEPs: differential expression proteins; MM: multiple myeloma

highly expressed and FGF–4, TACE, PIGF, VEGF–D, BMP–4, NGF R, BCMA, ANG–1, CD58, and TPO were lower expressed in the BiTe treatment group (Fig. 4, A). The expression of MMP–8, sFRP–3, IGF – 1R, TRAIL R1, IL–18 BPa, IL–29, E – Selectin, IFNab R2, MCSF, WIF–1, ErbB3, IGF–2R, OPN, Adiponectin, Ferritin, LIF, and IGFBP–6 were significantly higher in BiTe group than in PI-based group. And the level of Cathepsin B, IL–23, TIMP–2, ANGPTL3, bIG–H3, Kallikrein 14, IL–17, TGFb2, Eotaxin, Thrombomodulin, JAM–A, and Cadherin–4 in BiTe group were significantly decreased compare to PI-based group (Fig. 4, B). Besides, patients treated with anti-CD38 mAbs exhibited higher expression of Adiponectin, VEGF, MIF, sFRP–3, MMP–8, PIGF, IL–29, MMP–9, CD155, I–TAC, IL–18 BPa, ESAM, LIF, and MCSF, as well as lower expression of IL–2 alpha, TGFb RIII, Fractalkine, Osteoactivin, ANGPTL3, TIMP–, CD23, P–adherin, Cripto–1, AgRP, and Thrombomodulin than those received PI-based therapy (Fig. 4C).

### Functional enrichment analysis of DEPs and identification of hub-protein

Due to the significant difference in the rate of severe COVID-19 between the BiTe group and the PI-based group, we focused on the function of the DEPs between

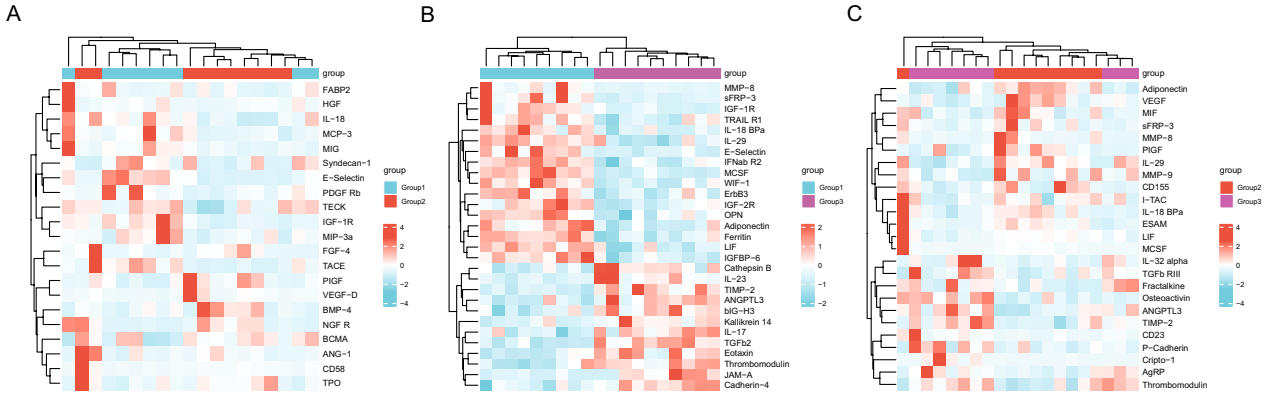


**Fig. 4** Hierarchical clustering analysis of DEPs among three treatment groups in MM patients. Group 1: MM patients treated with BCMA-CD3 bispecific antibody therapy; Group 2: MM patients treated with anti-CD38 mAbs therapy; Group 3: MM patients treated with PI-based therapy. **A** Results of hierarchical clustering analysis between MM patients treated with BiTe therapy and anti-CD38 mAbs therapy. **B** Results of hierarchical clustering analysis between MM patients treated with BiTe therapy and PI-based therapy. **C**. Results of hierarchical clustering analysis between MM patients treated with anti-CD38 mAbs therapy and PI-based therapy. Red, greater expression. Blue, less expression. DEPs: differential expression proteins; MM: multiple myeloma

these two groups. Functional analysis of DEPs between the two groups was further performed. Gene Ontology (GO) analysis includes three subtypes: biological process (BP), cellular component (CC), and molecular function (MF). The top 3 GO terms related to BP of DEPs between the BiTe group and PI-based group were primarily associated with the regulation of myeloid leukocyte differentiation (p adj: 0.00071), leukocyte migration (p adj: 0.00071), and granulocyte migration (p adj: 0.0010). There is a significant correlation in collagen-containing extracellular matrix (p adj: 0.00021), secretory granule lumen (p adj: 0.028), and cytoplasmic vesicle lumen (p adj: 0.028) in relation to CC. In addition, the terms of MF were mainly involved in cytokine activity (p adj: 5.57E-09), receptor ligand activity (p adj: 6.99E-08), and signaling receptor activator activity (p adj: 6.99E-08). KEGG pathway analysis showed that the DEPs between the BiTe group and PI-based group were mainly enriched in pathways of cytokine-cytokine receptor interaction (p adj: 5.21E-06), rheumatoid arthritis (p adj: 0.0063), and JAK-STAT signaling pathway (p adj: 0.026) (Fig. 5). Analysis of the DEPs enriched in cytokine-cytokine receptor interactions and JAK-STAT signaling pathways showed that in both pathways LIF is the most correlated with other DEPs and thus may play a key role (Fig. 6). The expression level of LIF protein in the BiTe group was higher than that in the anti-CD38 mAbs group ($p = 0.039$) (Fig. 7A). Compared to the PI-based group, the levels of LIF protein were significantly higher in both the BiTe group ($p < 0.001$) and anti-CD38 mAbs group ($p = 0.03$), with the BiTe group having the highest levels of LIF protein (Fig. 7A). Considering that COVID-19 infection itself may have an effect on LIF protein expression, LIF expression in total severe and non-severe COVID-19 patients across the three groups was analyzed, and found that there was no statistically significant difference in LIF expression between severe and non-severe patients ($p = 0.6$) (Fig. 7B). Besides, the GO and KEGG enrichment analysis of DEPs between the BiTe group and anti-CD38 mAbs group, and between the anti-CD38 mAbs group and PI-based group were shown in in Supplementary Figure S1.

## Discussion

In this study, for the first time, human protein microarrays were used to detect the expression of protein molecules after COVID-19 infection in three groups of MM patients treated with BiTe, anti-CD38 mAbs, and PI-based regimens. These findings provided insights into the mechanisms underlying the development of severe COVID-19 in MM patients receiving BiTe and laid the foundation for developing potential prognostic markers and therapeutic targets.

Our findings revealed that BiTe therapies were associated with a higher incidence of severe COVID-19. Both PCA and clustering analysis showed significant differences in protein expression profiles between the BiTe group and PI-based group. GO analysis showed that DEPs between the BiTe group and PI-based group mainly played molecular functions such as cytokine activity, receptor ligand activity, and signal receptor activator activity, and were mainly involved in the regulation of myeloid leukocyte differentiation, leukocyte migration, and granulocyte migration. Further KEGG pathway analysis showed that these DEPs were mainly enriched in pathways of cytokine-cytokine receptor interaction and JAK-STAT signaling pathway. Leukemia inhibitory factor

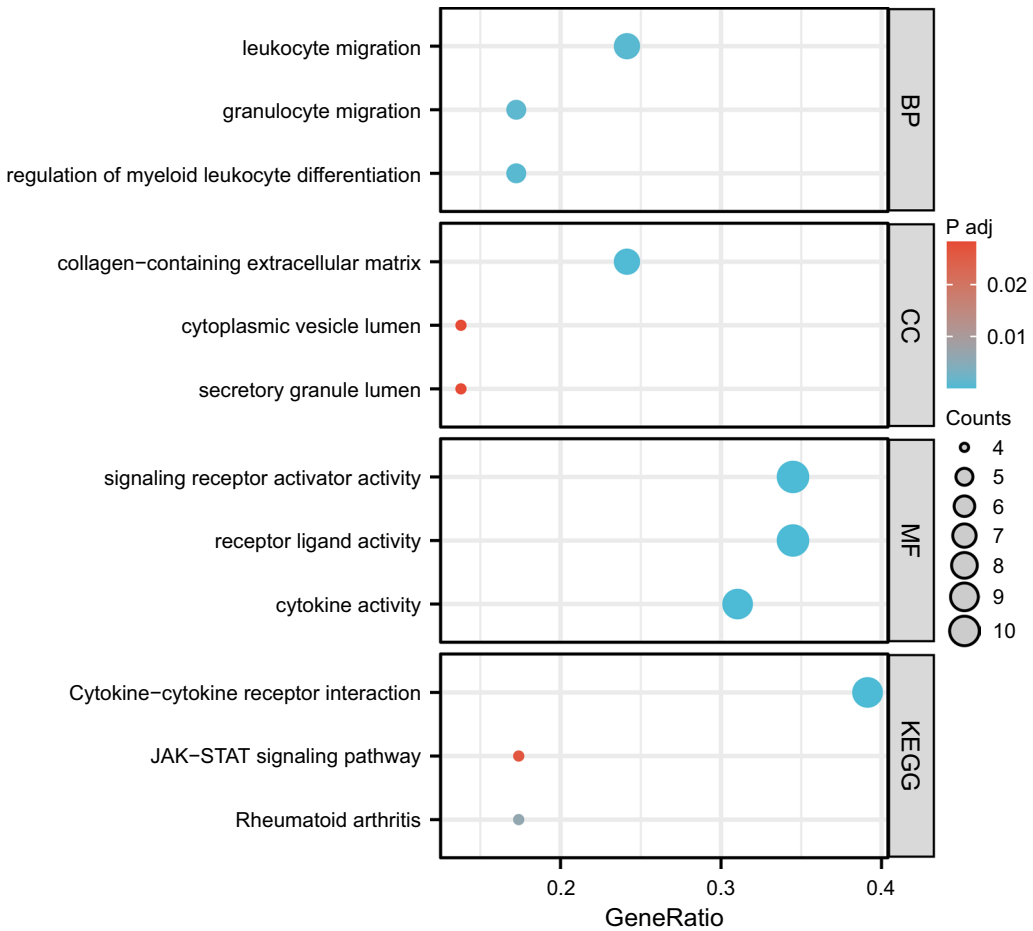


**Fig. 5** GO and KEGG enrichment analysis of DEPs between the BiTe group and the PI-based group. GO analysis includes three subtypes: BP (biological process), CC (cellular component), and MF (molecular function). Dot sizes represent counts of enriched DEPs, and dot colors represent p adjusted values. GO, Gene Ontology; KEGG, Kyoto Encyclopedia of Genes and Genomes; DEPs: differential expression proteins

(LIF) is the most correlated with other DEPs and thus may play a key role in both enriched pathways, and the level of LIF protein was highest in the BiTe group.

The higher incidence of severe COVID-19 in the BiTe group can be attributed to the profound immunomodulatory effects of BiTe therapy. BiTe binds simultaneously to BCMA expressed predominately on myeloma cells and to CD3 on T cells, bridging both cell types, resulting in T cell activation and thereby tumor cell lysis [7]. While effective in targeting MM cells, this mechanism can lead to excessive immune activation, potentially causing cytokine release syndrome (CRS) and other immune dysregulation syndromes, which could exacerbate the severity of COVID-19 infections. Since anti-CD38 mAbs and PI therapies involve less or no overactivation of the immune response, the possibility of causing a cytokine storm is much lower than that of BiTe treatment. This may account for the higher rate of severe COVID-19 in BiTe treated MM patients.

Convincingly, in our study, among the identified DEPs between the BiTe group and PI-based group, proteins involved in robust inflammatory and immune activation such as LIF, TRAIL R1, IL-18 BPa, IL−29, IFNab R2, MCSF, IGF−2R, OPN, and E-selectin were elevated in the BiTe group. Proteins involved in endothelial function and coagulation, such as ANGPTL3, bIG−H3, TGFb2, Thrombomodulin, and JAM−A were found to be lower in the BiTe group, potentially contributing to the coagulopathy and vascular complications seen in severe COVID-19 cases. These DEPs were mainly enriched in pathways of cytokine-cytokine receptor interaction and JAK-STAT signaling pathway. Much evidence suggests that cytokine storm is associated with the severity and mortality of COVID-19 patients [8]. Several studies have shown that the cytokine-cytokine receptor interaction pathway is an important pathway for the occurrence of cytokine storms and plays a critical role in the response to SARS-CoV-2 infection [9].

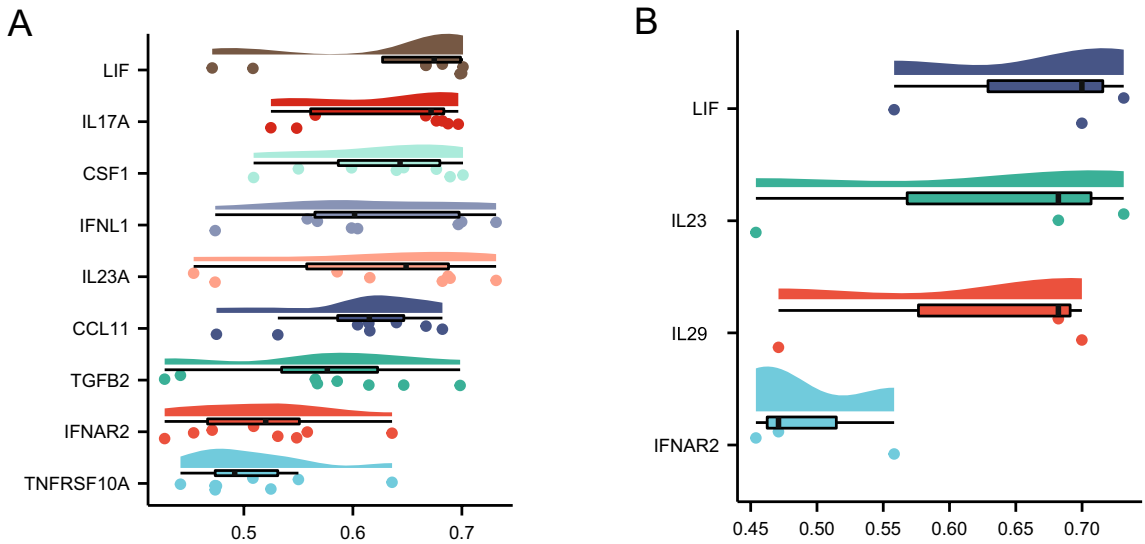


**Fig. 6** Hub protein identification of cytokine-cytokine receptor interactions and JAK-STAT signaling pathways. The x axis indicates the semantic similarity between the protein and other proteins, and the y axis indicates the corresponding protein, with the top protein indicating the key protein. **A** Enriched proteins in the cytokine-cytokine receptor interactions pathway. **B** Enriched proteins in the JAK-STAT signaling pathway

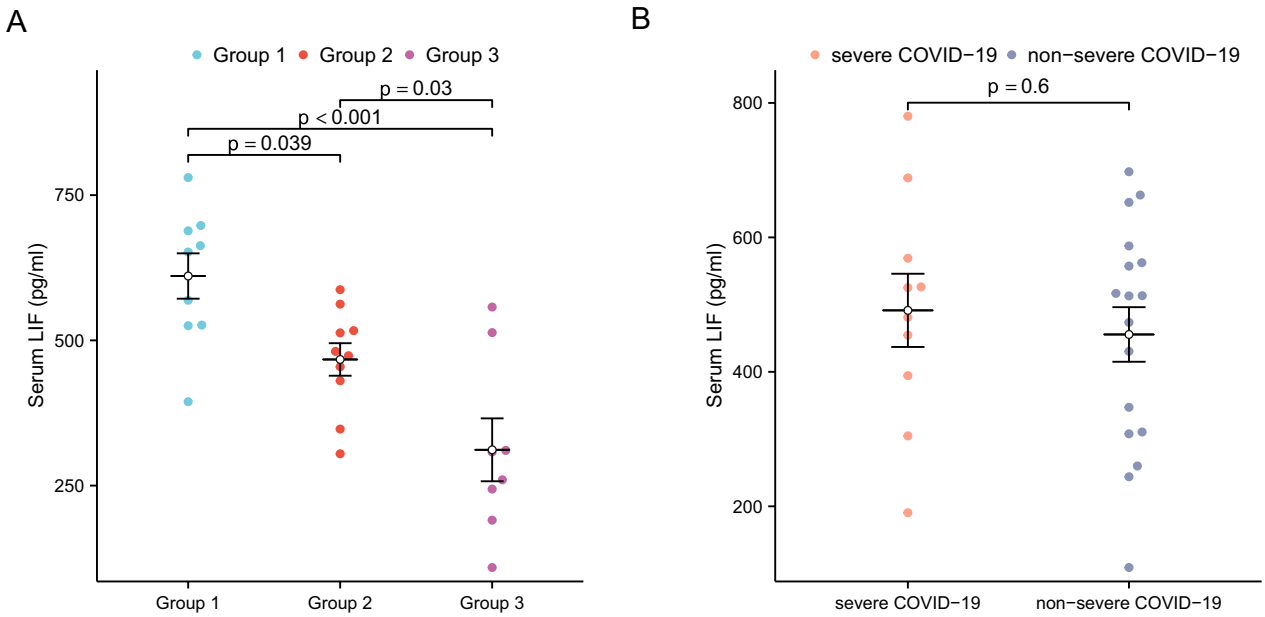


**Fig. 7** Expression levels of the hub protein LIF in MM patients with different treatment groups and MM patients with different severity of COVID-19. Group 1: MM patients treated with BCMA-CD3 bispecific antibody therapy; Group 2: MM patients treated with anti-CD38 mAbs therapy; Group 3: MM patients treated with PI-based therapy. **A** Plasma LIF level in MM patients treated with bispecific antibody therapy, anti-CD38 mAbs therapy, and PI-based therapy. **B** Plasma LIF level in MM patients infected with severe and non-severe COVID-19. LIF: leukemia inhibitory factor; MM: multiple myeloma

Moreover, the JAK-STAT signaling pathway is one of the critical cellular signaling pathways involved in the inflammatory response [10]. Studies have found JAK-STAT signaling pathway may as a central player of COVID-19, which is remarkably activated in patients with severe COVID-19 [11]. Inhibition of JAKs exerts potent anti-cytokine impacts in individuals with SARS-CoV-2 infection. Results have demonstrated that ruxolitinib improves pulmonary function in approximately 85% of COVID-19 patients with severe pulmonary

disease [12]. Baricitinib, a JAK inhibitor, received the FDA's emergency use authorization for the treatment of patients with severe COVID-19. The results of a randomized clinical trial in hospitalized COVID-19 individuals indicated that baricitinib along with remdesivir resulted in decreased hospitalization period and accelerated recovery time in patients receiving high-flow oxygen or noninvasive ventilation compared to remdesivir alone [13]. The fact that DEPs in our study were predominantly enriched in cytokine-cytokine receptor interaction and JAK-STAT signaling pathway reinforces the idea: BiTe treatment predisposes to cytokine storm, which can lead to severe COVID-19 infection in MM patients.

Based on the results of Friend analysis, LIF plays a vital role in severe COVID-19 development in MM patients treated with BiTe by interacting strongly with other proteins in cytokine-cytokine receptor and JAK-STAT signaling pathways. LIF is a multi-functional cytokine of the interleukin-6 (IL-6) superfamily [14]. LIF signals through the heterodimeric glycoprotein 130/LIF receptor complex activating the JAK-STAT pathway, displaying a wide variety of important functions in many physiological and pathological processes [14]. It is well known that the positive relationship of elevated IL-6 levels with COVID-19 severity indicates its key role in the progression and pathogenesis of the COVID-19 disease [15]. However, little is known about the role of LIF molecules in COVID-19. A recent study showed that LIF will be induced in response to pro-inflammatory stress stimulation in macrophages, which contributes to the immune response to viral and bacterial infections [16]. Mendelian randomization (MR) analysis in Genome-Wide Association Studies (GWAS) result from Baranova A et al. found the severe course of SARS-CoV-2 would be expected when levels of LIF are higher [17]. Conversely, Metcalfe SM thought that LIF molecule could be used to treat severe COVID-19 since it was found to improve lung tissue resilience in animal models of pneumonia [18]. Our results observed LIF protein was expressed at the highest level in the BiTe group with more severe COVID-19, implying it may serve as a potential prognostic marker for severe COVID-19 in MM patients treated with BiTe. LIF inhibitors and JAK-STAT pathway inhibitors can be used to treat severe COVID-19 in MM patients.

Additionally, the latest international consensus on BiTe treatment-related infections indicates that bispecific antibody therapy presents an elevated risk of infections and severe conditions compared to conventional MM treatment regimens [19]. The guidelines emphasize that the most likely explanation is that BiTe treatment significantly activates T-cells, leading to CRS and immune system dysregulation, resulting in serious infections. This mechanism is not limited to COVID-19 but is also observed in various other serious infections. Studies have reported cases of reactivation of cytomegalovirus (CMV) and hepatitis B virus (HBV), severe bacterial infections such as pneumonia, sepsis, and urinary tract infections following BiTe treatment [20]. Fungal infections have also been noted, particularly aspergillosis and candidiasis, which can lead to challenging systemic disseminated infections [21]. In summary, understanding the specific mechanisms by which BiTe therapy leads to serious infections can help implement comprehensive infection monitoring and management strategies in MM patients, ultimately improving patient prognosis.

Our results should be interpreted with caution in light of some limitations. First, this was a small cohort from single center. However, our results could be further validated in a larger population since several BiTes have been approved in China. Second, the effect of COVID-19 infection itself on protein expression cannot be completely excluded. Besides, the exact cause and mechanism of LIF elevation induced by BiTe treatment have not been clarified, and further investigation is needed.

## Conclusions

In conclusion, our study highlights the different effects and mechanisms of different MM treatment regimens on the severity of COVID-19, with the BiTe therapy associated with a higher risk of severe outcomes. The main pathways involved in this process are cytokine-cytokine receptor interaction and the JAK-STAT signaling pathway. And LIF as a key molecule in these two pathways, could serve as a potential prognostic marker. These findings suggest that LIF and JAK-STAT pathway can serve as potential therapeutic targets for attenuating severe COVID-19 in MM patients treated with BiTe. Novel serious infections related to BiTes either by inflammatory cytokine storm or immune suppression should be greatly stressed for general concerns.

**Abbreviations**

| | |
|---|---|
| MM | Multiple myeloma |
| BiTe | Bispecific T-cell engagers |
| mAbs | Monoclonal antibodies |
| PI | Proteasome inhibitor |
| DEPs | Differentially expressed proteins |
| GO | Gene ontology |
| KEG | Kyoto encyclopedia of genes and genomes |
| LIF | Leukemia inhibitory factor |
| CAR T cells | Chimeric antigen receptor T cells |
| COVID-19 | Coronavirus disease 2019 |
| BCMA | B cell maturation antigen |
| R/R MM | Relapsed/refractory MM |
| SD | Standard deviation |
| IQR | Interquartile range |
| FDR | False discovery rate |
| PCA | Principal component analysis |
| R-ISS | Revised-international staging system |
| auto-HSCT | Autologous hematopoietic stem cell transplantation |
| PC | Principal components |

| | |
|---|---|
| BP | Biological process |
| CC | Cellular component |
| MF | Molecular function |
| CRS | Cytokine release syndrome |
| MR | Mendelian randomization |
| GWAS | Genome-Wide Association Studies |
| CMV | Cytomegalovirus |
| HBV | Hepatitis B virus |

## Supplementary Information

The online version contains supplementary material available at https://doi.org/10.1186/s12967-025-06140-y.

Additional file 1

**Acknowledgements**
We acknowledge all patients and healthy individuals participated in this study.

**Author contributions**
Ziping Li drafting and revision of the submitted article. Fujing Zhang and Xianghong Jin, sample collection and data analysis. Junling Zhuang design of the study and revision of the submitted article. The work reported in the paper has been performed by the authors, unless clearly specified in the text.

**Funding**
This work was supported by a Capital's Funds for Health Improvement and Research (Grant No. 2022–2-4013) and National High Level Hospital Clinical Research Funding (Grant No. 2022-PUMCH-B-048).

**Availability of data and materials**
All data are available upon request.

## Declarations

**Ethics approval and consent to participate**
This study was approved by the Institutional Medical Ethics Committee (No. I-23PJ229) at Peking Union Medical College Hospital.

**Consent for publication**
Not applicable.

**Competing interests**
All authors declare that they have no competing interest.

Received: 20 November 2024 Accepted: 15 January 2025
Published online: 26 January 2025

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